

SECTION A

1 **(short)** Solve the difference equation

$$x_{n+2} - 5x_{n+1} + 4x_n = 0$$

with initial values of $x_0 = 1$ and $x_1 = 5$.

[10]

2 **(short)**

(a) Using l'Hôpital's rule, find

$$\lim_{x \rightarrow 0} \left[\frac{\sin x - \sin^2 x}{x^2 - x + \sin^2 x} \right]$$

[5]

(b) Using the method of power series expansion, find

$$\lim_{x \rightarrow 0} \left[\frac{(\sin x - x)}{x^3} - \frac{1}{\cosh x} \right]$$

[5]

3 **(short)** For the matrix

$$A = \begin{bmatrix} 8 & 3 \\ 1 & 6 \end{bmatrix}$$

determine the eigenvalues and eigenvectors.

[10]